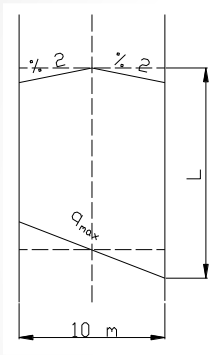
INS3441

HIGHWAY ENGINEERING

Prof. Halit ÖZEN, PhD.

HORIZANTAL CURVES

The application of the superelevation



$$p' = \frac{V^3}{46{,}7 * R * L} - \cancel{\frac{V * q}{46{,}7 * L}}$$

neglected

$$p'_{max} = 0.6 \text{ m/sec}^3$$

$$L(m) = 0{,}0354 * \frac{V^3 \left(\frac{km}{h}\right)}{R\ (m)} \geq 45\ (m)$$

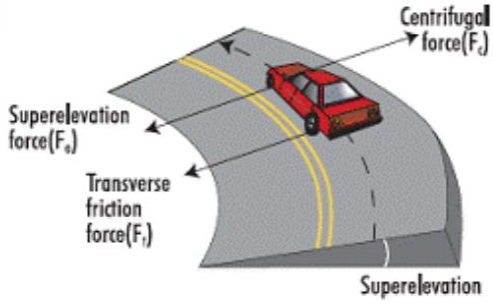
HORIZANTAL CURVES







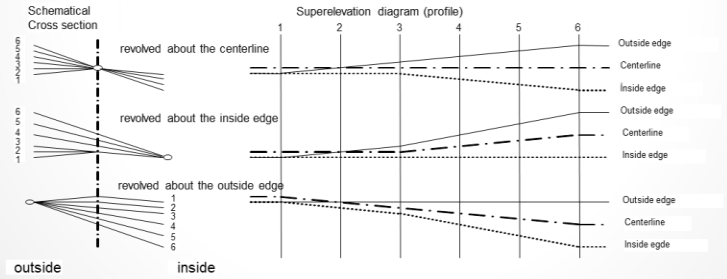
HORIZANTAL CURVES



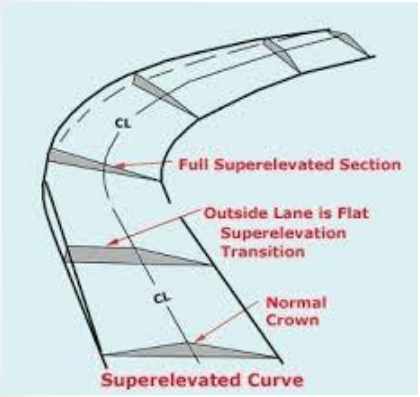
HORIZANTAL CURVES

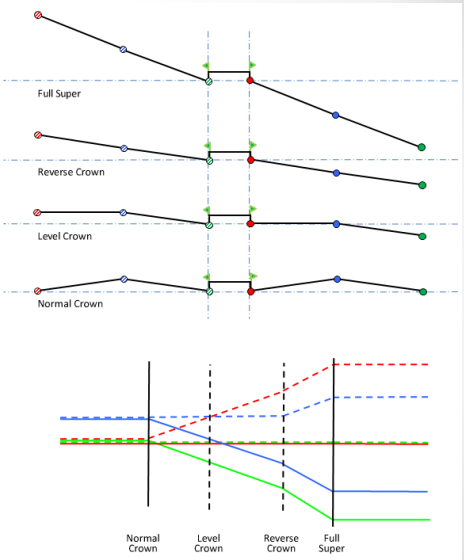
Implementing the change of superelevation through the length

- Revolve traveled way with normal cross slopes about **the centreline** profile
- Revolve traveled way with normal cross slope about the **outside-edge** profile
- Revolve traveled way with normal cross slope about the **inside-edge** profile



HORIZANTAL CURVES

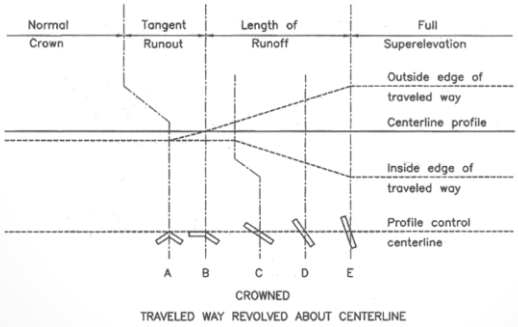




HORIZANTAL CURVES

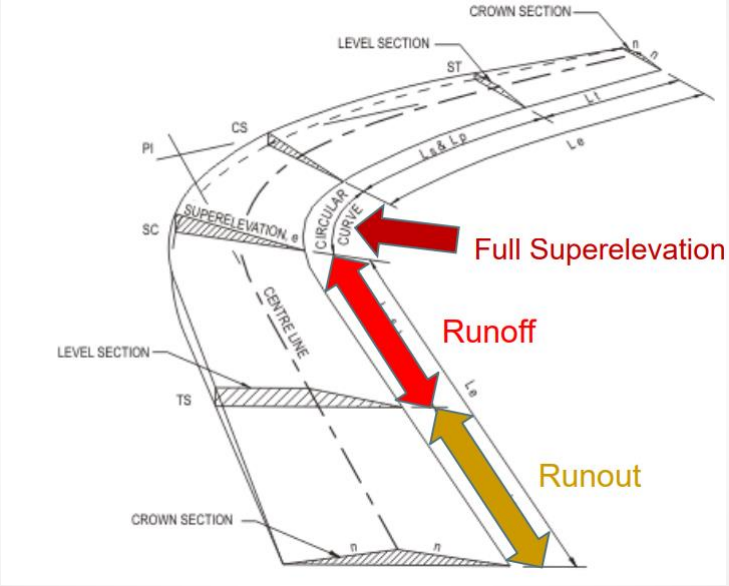
Superelevation Transitions Consists of Runoff and Tangent Runout sections.

- **Runoff:** length of roadway needed to accomplish a change in outside lane cross slope from zero to full
- **Runout:** length of roadway needed to accomplish a change in outside lane cross slope from normal rate to zero



from AASHTO's *A Policy on Geometric Design of Highways and Streets 2001*

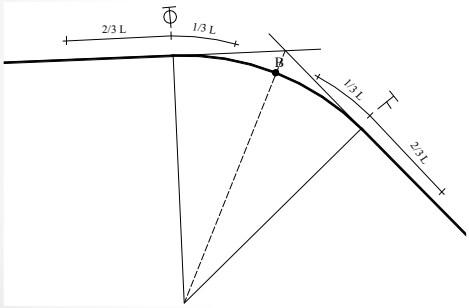
HORIZANTAL CURVES



HORIZANTAL CURVES

The cross slope is started to be changed 2/3*L_d before the BC ($\overline{\Phi}$) and ended at 1/3* L_d after the BC ($\overline{\Phi}$).

On the other side of the curve full superelevation started to be changed 1/3*L_d before EC ($\overline{\digamma}$) and chaged to cross slope 2/3*L_d after EC ($\overline{\digamma}$)



$$DB_{\overline{\Phi}}(km) = \overline{\Phi}(km) - \frac{2}{3} * L$$

$$DS_{\overline{\Phi}}(km) = \overline{\Phi}(km) + \frac{1}{3} * L$$

$$DB_{\overline{\digamma}}(km) = \overline{\digamma}(km) - \frac{1}{3} * L$$

$$DS_{\overline{\digamma}}(km) = \overline{\digamma}(km) + \frac{2}{3} * L$$